Design Document: MtG Deckbuilding Assistant

1. Purpose

The goal of the project is to create a tool that allows the user to shortlist a selection of Magic the Gathering cards, evaluate each one through a structured, weighted scoring system, and rank them from strongest to weakest. It is meant to support comparative evaluation — helping the user decide how strong one card is relative to every other card in the same list — rather than to obtain absolute values or free-form notes

A companion capability builds a local reference library of card data (name, artwork, and rules text) sourced from the Scryfall public database, so the user can look up and preview real cards while building their evaluations, instead of typing everything from scratch.

2. Core Concepts

2.1 Cards

A Card is the basic unit being evaluated — typically a single card. Each Card has:

- A name (optionally matched against the card reference library for autocomplete and preview)
- One or more tags
- Zero or more references to other cards in the same collection
- A "frozen" state and, a "locked" state and an “free” state (see §5)

A collection may hold up to 200 cards at a time.

2.2 Tags and Sub-Tags

A tag represents one dimension of a card's value (e.g. a role or effect the card provides). Every card always carries a special Mana tag, plus up to nine additional user-defined tags (ten tags total).

Each tag breaks down into a small set of sub-tags, which the user scores individually:

- Mana tag: broken into *Cost* (how expensive the card is to use, scored 1–15) and *Burden* (how many colored pips are needed to use the card, scored 1–10)
- Every other tag: broken into *Speed*, *Resilience*, and *Impact* (each scored 1–10)

Cost,Burden and Speed are sub-tags where a higher input value is intentionally worth less: a cheaper card(low Cost or Colour Burden) or a faster effect (low Speed) is treated as stronger, so these two sub-tags' contributions are calculated as "maximum minus the entered value" rather than added directly. Resilience and Impact are added at face value — higher is simply better.

2.3 Tag Groups

Non-Mana tags can optionally be grouped together under a shared label. Grouping expresses "these tags reinforce each other" rather than being fully independent sources of value. A tag left ungrouped is still treated as belonging to its own single-tag group — grouping only matters in how the resulting groups are weighted (see §3).

2.4 References

A card can reference up to 200 other cards in the same collection, each with a weight from −10 to +10. A reference expresses a relationship — typically synergy or anti-synergy — between two cards, and contributes a bonus (or penalty) to the score of each side.

References are always two-way: creating a reference from Card A to Card B automatically creates a matching reference back from B to A, using the same weight. Removing the link on either side removes it on both. Because the bonus each side receives is calculated using its own score relative to the other's, the two sides of the same reference usually contribute different amounts to each card's total.

3. Scoring Model

3.1 Tag Score

Each tag has its own score:

- Mana tag score = a fixed baseline (27, can be changed later) minus Cost, minus Colour Burden. Lower Cost and lower Colour Burden both raise this score
- Other tag score = (a fixed baseline 11 minus Speed) + Resilience + Impact. Lower Speed raises this score; higher Resilience and Impact both raise it

3.2 Group Weighting

Non-Mana tags are organized into groups (see §2.3). Each group's combined score is added to the total score, but weighted by rank: the single highest-scoring group counts at full weight, the next-highest at half weight, the next at a quarter, and so on, halving each step down. This rewards cards whose value is concentrated in one strong, coherent group of tags/abilities, rather than spread across many unrelated ones (a card that can do x and y is better than one that can do x, y or z)

3.3 Reference Bonus

For each reference a card holds to another card, a bonus is calculated from:

- the other card’s tag score, squared
- divided by the card’s own tag score
- multiplied by the reference's weight
- and scaled down by a small fixed factor (x0.01)

A negative reference weight produces a negative bonus (a penalty) using the same calculation. All of a card's individual reference bonuses are summed to produce its total reference bonus. This calculation only ever looks at the two cards' own tag scores — never at their reference bonuses — so long chains or mutual references between cards can never spiral or loop.

3.4 Total Score

A card's total score is its tag score (Mana score plus weighted groups) plus its total reference bonus. The full collection is ranked by total score, highest first, with ties broken by alphabetic order

4. Working With the Collection

4.1 Ranked List

All (non-frozen) cards are displayed together, ranked by total score. The list can instead be sorted by name, by tag score or by reference bonus; each column can be sorted ascending or descending.

4.2 Filtering and Search

The user can search the list by name (matching any part of the name, not just the start) and can hide or show only all cardos that carry a particular tag, to focus on a subset of the collection. Filters don't affect scoring — only what's currently shown.

4.3 Inspecting Relationships

Selecting a card highlights every other card it has a reference relationship with (in either direction, since references are always mirrored). When exactly one card is selected, each related card's entry additionally shows the specific bonus that relationship contributes to the selected card's score, rather than just its overall total.

4.4 Freezing Cards

A card can be "frozen," which sets it aside into a separate holding area outside the main ranked list. A frozen card:

- Is excluded from the ranking, from tag counts used for filtering, and from being offered as a reference target for other cards.
- Keeps all of its own tags, groups, and references intact, and remains fully editable.
- Has any reference connecting it to a live card treated as contributing zero bonus on both sides, for as long as it stays frozen.

This is meant for setting a card aside for later reconsideration without deleting its data or letting it continue to influence the live ranking.

4.5 Locking the Editing Panel

Normally, clicking a different card in the ranked list loads that card into the editing panel. Locking the editing panel keeps the currently open card's name, tags, and references in place even as different rows are clicked — only the artwork/rules-text preview updates to match whatever was clicked. Saving while locked keeps the same card open afterward rather than clearing the panel, which supports repeatedly refining one card's tags while glancing at other cards for comparison.

4.6 Copying a Tag Between Cards

A tag (its name and all of its sub-tag scores) can be copied from the card currently being edited onto any other card in the ranked list, with a confirmation step before it's applied. This is a convenience for giving several cards an identical tag without re-entering its values by hand.

5. Card Reference Library

A separate preparation step builds a local library of card data — name, artwork, and rules text — from a public card-game database, covering both single-sided and two-sided cards (for two-sided cards, the two halves' rules text are combined and the front side's artwork is used). Cards missing artwork or rules text are excluded from the library.

Within the main tool, typing an card's name searches this library and offers matching suggestions, favoring names that start with what's been typed over names that merely contain it. Once the name exactly matches a library entry, that card's artwork and rules text are shown alongside the editing panel for reference, and the artwork can be enlarged into a new window for a closer look. If the library isn't available, the name field still works as free text; only the autocomplete and preview are unavailable.

6. Saving and Organizing Collections

Every change is saved automatically as it's made — there is no separate save step to remember. The user can start a new, empty collection, open a previously saved one, or save the current

collection under a new name at any time; whichever collection was open most recently reopens automatically the next time the tool is started.

7. Display Preferences

The user can choose from a small set of preset window sizes (matching common screen resolutions), which resizes and re-centers the window and scales the interface accordingly. This preference is remembered between sessions. Changing it while already running only reliably applies in full the next time the tool starts.

8. Limits Summary

Item	Limit
Cards per collection	200
Tags per card	1–10 (mana tag always counts as one)
Cost (Mana tag)	1–15
Burden (Mana tag)	1–10
Speed / Resilience / Impact (other tags)	1–10 each
References per card	0–200
Reference weight	−10 to +10
Card-name suggestions shown while typing	up to 10